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Inference of progressively censored competing risks data from Kumaraswamy distributions

Liang Wang

Xidian University, School of Mathematics and Statistics, No. 2 Taiba Nanlu, Xi'an, Shaanxi Province 710071, PR China

HIGHLIGHTS

- Inference for different and common competing risks parameters is considered.
- Maximum likelihood estimators are established.
- Confidence intervals are proposed via observed fisher information matrix.
- Bayesian estimates are derived via MCMC sampling method.
- Data example on football game proportion time is illustrated.

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ABSTRACT

A competing risks model based on Kumaraswamy distribution is discussed under progressive censoring. When the latent lifetime model of failure causes features different and common parameters, maximum likelihood estimates for unknown parameters are established where the existence and uniqueness of the estimates are provided, and the approximate confidence intervals are also constructed via the observed fisher information matrix. Moreover, Bayes estimates and associated highest posterior density credible intervals are also obtained based on Monte-Carlo Markov chain sampling methods. In addition, to test the equivalence of parameters between the competing risks, likelihood ratio test is also proposed. Finally, simulation studies and real-life example are presented for illustration purpose.

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1. Introduction and notation

Due to the complexity of internal structure and external environment, the failure of units is commonly yielded by multiple causes of failure which in some sense compete with each other in practice. In literature, these failure causes are called as competing risks where the observed failure data consists of a failure time and an indicator denoting the cause of failure. Competing risks have received wide attention and are frequently encountered in various fields such as survival analysis, lifetime study and reliability among others. For instance, Cramer and Schmiedt [1] discussed Lomax competing risks model under progressive Type-II censoring. Pareek et al. [2] studied Weibull competing risks model under incomplete data. Hyun et al. [3] considered the proportional hazards model for competing risks data with missing cause of failure. Balakrishnan and Han [4] and Pascual [5] discussed the sampling plan problems in accelerated life test when the failure cause of competing risks is exponentially and lognormally distributed, respectively.

E-mail address: wangl@xidian.edu.cn.

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Besides competing risks, due to the restriction of cost and time, it is impossible to test products in a complete life cycle and censoring scheme (CS) is introduced in practice. Generally speaking, censoring implies that exact failure times are known for only a portion of the units under study. There are several types of CSs, and the most common ones are Type-I censoring and Type-II censoring, where n units are tested and the experiment is terminated after a prefixed time T or a prefixed failure number $m(< n)$. With the improvement of technique and manufacture, however, products feature characteristic of high reliable and long span, which make experiments cannot obtain lifetime information of products in a short period. In order to take a more efficient way to obtain failure data, progressive censoring scheme (PCS) is further proposed in practice. In general, PCS includes progressive Type-I censoring and progressive Type-II censoring respectively, where it allows engineers to remove survival units from testing at various stages due to prefixed time or prefixed number of failure. It is noted that PCS is also widely used in reliability analysis and life testing, and is more flexible and efficient than conventional Type-I and Type-II CSs. For details, one can refer to recent review paper by Balakrishnan [6] and monograph of Balakrishnan and Aggarwala [7] as well as the references therein.

Let X be a random variable from the Kumaraswamy distribution with two positive shape parameters $\alpha > 0$ and $\beta > 0$, of which the cumulative distribution function (CDF) and the probability density function (PDF) are given respectively by

$$\begin{aligned} &F(x;\alpha,\beta) = 1-[1-x^{\alpha}]^{\beta},\\ &f(x;\alpha,\beta) = \alpha\beta x^{\alpha-1}[1-x^{\alpha}]^{\beta-1},\ 0<x<1. \end{aligned} \tag{1}$$

The model (1) is proposed by Kumaraswamy [8] for describing hydrological random variables such as daily rainfall and daily stream flow and is introduced as an alternative model to the beta distribution. According to Nadarajah [9], this distribution is used in the hydrological sciences because it is deemed as a better alternative to the beta distribution. The Kumaraswamy distribution is a very flexible model whose density function can be unimodal, uniantimodal, increasing, decreasing or constant depending on the values of its parameters. It can be shown that the Kumaraswamy distribution has the same basic shape properties of the beta distribution: $\alpha > 1$ and $\beta > 1$ (unimodal); $\alpha < 1$ and $\beta < 1$ (uniantimodal); $\alpha > 1$ and $\beta \leq 1$ (increasing); $\alpha \leq 1$ and $\beta \leq 1$ (decreasing); $\alpha = \beta = 1$ (constant). Recently, the Kumaraswamy distribution has received considerable attention in literatures and practice due to its beta-type and better than the beta distribution by its explicit expression of quantile function (see, e.g., Jones [10]), and this distribution also shows well goodness-of-fit for many natural phenomenons such as hydrological data, atmosphere temperatures among others (see, e.g. Kumaraswamy [8] and Nadarajah [9]). Meanwhile, the Kumaraswamy can be also employed to approximate many distributions such as uniform, triangular, or almost any single model distribution on the choice of parameters α and β, and can also reproduce results of the truncated normal or beta distributions (see, e.g, Kumaraswamy [8] and Ponnambalam et al. [11]). Motivated by such reasons as mentioned above, inference for the Kumaraswamy distribution is of considerable interest and of practical significance in many practical situations, this paper studies estimation for Kumaraswamy parameters when there are competing risks involved. Under progressive Type-II censoring, point and interval estimates are investigated under classic and Bayesian procedures, respectively.

The rest of this paper is organized as follows. Model description and the likelihood function are introduced in Section 2. Under classic and Bayesian approaches, point and interval estimates are presented in Sections 3 and 4 respectively, when the latent failure distributions have different and common model parameters. Some testing problems are proposed in Section 5. Section 6 presents some simulation studies and real-life example. Finally, some concluding remarks are provided in Section 7.

2. Model description and likelihood function

Consider a progressive Type-II censoring lifetime testing with n identical units, of which the corresponding lifetimes are described by independent and identically distributed (i.i.d.) random variables $X_1, X_2, \ldots, X_n$. Suppose there are two causes of failure which are known in advance, one has $X_i = \{X_{1i}, X_{2i}\}$ for $i = 1, 2, \ldots, n$, where X_{ki}, $k = 1, 2$ denotes the latent failure time of the ith unit under kth cause of failure. In order to avoid some identifiability issues we assume independence of the causes of failure, and the progressively Type-II censored competing risks model can be described as follows: suppose n independent units are put in test and a prefixed CS $R = (r_1, r_2, \ldots, r_m)$ is also given. When first failure time, say $X_{1:m:n}$, has occurred, the cause of failure δ_1 is recorded and r_1 surviving units are randomly removed from the $n-1$ remaining units. When second failure time, say $X_{2:m:n}$, has occurred, the cause of failure δ_2 is recorded and r_2 surviving units are randomly removed from the $n-r_1-2$ remaining units. Proceeding the procedure until the mth failure, say $X_{m:m:n}$, the cause of failure δ_m is recorded and all the remaining $r_m = n-m-r_1-\cdots-r_{m-1}$ surviving units are removed. Therefore, the progressively Type-II censored competing risks sample is given by

$$[X_{1:m:n}, \delta_1, r_1], [X_{2:m:n}, \delta_2, r_2], \ldots, [X_{m:m:n}, \delta_m, r_m]. \tag{2}$$

It is seen that when indicator $\delta_i = 1$, the ith failure is yielded due to cause one and $\delta_i = 2$ otherwise. Moreover, let

$$\mathbb{1}(\delta_i = 1) = \begin{cases} 1, & \delta_i = 1;\\ 0, & \text{otherwise.} \end{cases} \quad \text{and} \quad \mathbb{1}(\delta_i = 2) = \begin{cases} 1, & \delta_i = 2;\\ 0, & \text{otherwise} \end{cases}$$

then the joint PDF of (2), namely $L(x)$, can be expressed as

$$L(x) = c\prod_{i=1}^{m}\prod_{k=1}^{2}[f_k(x_{i:m:n})\overline{F}_{3-k}(x_{i:m:n})]^{\mathbb{1}(\delta_i=k)}[\overline{F}_1(x_{i:m:n})\overline{F}_2(x_{i:m:n})]^{r_i}, \tag{3}$$

where $f_k(\cdot)$ and $F_k(\cdot) = 1-\overline{F}_k(\cdot)$, $k = 1, 2$ denote the PDF and CDF of the kth cause of failure, and $c = n\prod_{i=1}^{m-1}[n-\sum_{j=1}^{i}(r_j+1)]$.

Suppose that kth competing risk X_{ki}, $k = 1, 2, i = 1, 2, \ldots, n$ are i.i.d. random variables from the Kumaraswamy distribution (1) with parameter (α_k, β_k), the likelihood function of β_1, β_2 and α_1, α_2, say $L(\beta_1, \beta_2, \alpha_1, \alpha_2; x)$, is given by

$$\begin{aligned} L(\beta_1, \beta_2, \alpha_1, \alpha_2; x) &= c\alpha_1^{m_1}\alpha_2^{m_2}\beta_1^{m_1}\beta_2^{m_2}\left[\prod_{i=1}^{m}x_{i:m:n}^{\mathbb{1}(\delta_i=1)}\right]^{\alpha_1-1}\left[\prod_{i=1}^{m}x_{i:m:n}^{\mathbb{1}(\delta_i=2)}\right]^{\alpha_2-1} \\ &\cdot\prod_{i=1}^{m}[1-x_{i:m:n}^{\alpha_1}]^{(r_i+1)\beta_1-\mathbb{1}(\delta_i=1)}\cdot\prod_{i=1}^{m}[1-x_{i:m:n}^{\alpha_2}]^{(r_1+1)\beta_2-\mathbb{1}(\delta_i=2)}, \end{aligned} \tag{4}$$

where $m_k = \sum_{i=1}^{m}\mathbb{1}(\delta_i = k)$, $k = 1, 2$ satisfying $m_1 + m_2 = m$ describes the number of failure due to kth cause of failure in m observations.

Remark 1. It is seen that the competing risks model proposed in this paper is actually constructed in Cox's latent failure times model formulation. The situation may can be analyzed as well by using a mixture model. Instead of considering competing risks above, one may assume that the underlying lifetime model is a mixture of two Kumaraswamy distributions as

$$F(x) = pF(x; \alpha_1, \beta_1) + (1-p)F(x; \alpha_2, \beta_2),\ 0 < x < 1,$$

where the mixture probability p is unknown, and the data (2) can be viewed as a progressively Type-II censored sample from the population distribution function $F(x)$ with parameters β_1, β_2, α_1, α_2 and p. As in above model, the estimation procedure is based on a progressively Type-II censored sample where additional information on the cause of failure is available. Besides above mentioned mixture distribution, the cause specific hazard function formulation model is also a useful procedure to analyze competing risks data in practice where different hazard functions are assumed for causes of failure, see, for example, the works of Caplan et al. [12], Cheng and Fine [13] for more details.

3. Frequentist estimation

The maximum likelihood estimates (MLEs) and approximate confidence intervals (ACIs) are provided for unknown parameters in this section when there are different and common model parameters.

3.1. Maximum likelihood estimation with different parameters

From (4), the log-likelihood function of β_1, β_2 and α_1, α_2, say $\ell(\beta_1, \beta_2, \alpha_1, \alpha_2)$, is given by

$$\begin{aligned} \ell(\beta_1, \beta_2, \alpha_1, \alpha_2) &= \sum_{k=1}^{2}m_k\ln\alpha_k + \sum_{k=1}^{2}m_k\ln\beta_k + \sum_{k=1}^{2}\sum_{i=1}^{m}\alpha_k\mathbb{1}(\delta_i=k)\ln x_{i:m:n} \\ &+ \sum_{k=1}^{2}\sum_{i=1}^{m}[(r_i+1)\beta_k - \mathbb{1}(\delta_i=k)]\ln(1-x_{i:m:n}^{\alpha_k}) + C, \end{aligned} \tag{5}$$

where $C = \ln c - \sum_{i=1}^{m}\ln x_{i:m:n}$.

Theorem 1. *Suppose the latent competing risks failure times come from Kumaraswamy distributions with parameters (α_1, β_1) and (α_2, β_2), for $m_k > 0$ and given α_k, $k = 1, 2$, the MLE of β_k exists and is given by*

$$\dot{\beta}_k = -\frac{m_k}{\sum_{i=1}^{m}(r_i+1)\ln(1-x_{i:m:n}^{\alpha_k})} \quad k = 1, 2.$$

Proof. See Appendix A. □

Theorem 2. *Suppose the latent competing risks failure times come from Kumaraswamy distributions with parameters (α_1, β_1) and (α_2, β_2). For $m_k > 0$, $k = 1, 2$, the MLE of α_k exists and is unique which can be obtained from following equation*

$$\dot{H}(\alpha_k) = \frac{m_k}{\alpha_k}\left[1 + \dot{H}_1(\alpha_k) + \frac{\dot{H}_2(\alpha_k)}{\dot{H}_3(\alpha_k)}\right] = 0, k = 1, 2, \tag{6}$$

where

$$\dot{H}_1(\alpha_k) = \frac{1}{m_k}\sum_{i=1}^{m}\frac{\mathbb{1}(\delta_i = k)\ln x_{i:m:n}^{\alpha_k}}{1 - x_{i:m:n}^{\alpha_k}},$$
$$\dot{H}_2(\alpha_k) = \sum_{i=1}^{m}(r_i + 1)\frac{x_{i:m:n}^{\alpha_k}\ln x_{i:m:n}^{\alpha_k}}{1 - x_{i:m:n}^{\alpha_k}},$$
$$\dot{H}_3(\alpha_k) = \sum_{i=1}^{m}(r_i + 1)\ln(1 - x_{i:m:n}^{\alpha_k}).$$

Proof. See Appendix B. □

Denote the MLE of α_k, $k = 1, 2$ derived from (6) in Theorem 2 as $\hat{\alpha}_k$, then the MLE of β_k, namely $\hat{\beta}_k$, exists and can be expressed from Theorem 1 as

$$\hat{\beta}_k = -\frac{m_k}{\sum_{i=1}^{m}(r_i + 1)\ln(1 - x_{i:m:n}^{\hat{\alpha}_k})}, \quad k = 1, 2. \tag{7}$$

Remark 2. By construction $m_1 + m_2 = m > 0$, so that either $m_1 > 0$ or $m_2 > 0$. If there are no failures due to kth cause ($k = 1, 2$), i.e., $m_k = 0$, the observed competing data (2) provide no information about unknown parameters (α_k, β_k) and the associated MLEs do not exist.

3.2. MLEs under common parameter cases

In this subsection, MLEs for Kumaraswamy competing risks parameters are presented when common parameters are available.

• Inference for common parameters $\alpha_1 = \alpha_2 = \alpha$.

Theorem 3. *Suppose the latent competing risks failure times come from Kumaraswamy distribution with parameters (α, β_1) and (α, β_2), for $m_k > 0$, $k = 1, 2$ and given α, the MLE of β_k exists and is given by*

$$\ddot{\beta}_k = -\frac{m_k}{\sum_{i=1}^{m}(r_i + 1)\ln(1 - x_{i:m:n}^{\alpha})}, \quad k = 1, 2.$$

Proof. See Appendix C. □

Theorem 4. *Suppose the latent competing risks failure times come from Kumaraswamy distribution with parameters (α, β_1) and (α, β_2), for $m_1 > 0$, $m_2 > 0$, the MLE of α exists and is unique which can be obtained from following equation*

$$\ddot{H}(\alpha) = \frac{m}{\alpha}\left[1 + \ddot{H}_1(\alpha) + \frac{\ddot{H}_2(\alpha)}{\ddot{H}_3(\alpha)}\right] = 0, \tag{8}$$

where

$$\ddot{H}_1(\alpha) = \frac{1}{m}\sum_{i=1}^{m}\frac{\ln x_{i:m:n}^{\alpha}}{1 - x_{i:m:n}^{\alpha}},$$
$$\ddot{H}_2(\alpha) = \sum_{i=1}^{m}(r_i + 1)\frac{x_{i:m:n}^{\alpha}\ln x_{i:m:n}^{\alpha}}{1 - x_{i:m:n}^{\alpha}},$$
$$\ddot{H}_3(\alpha) = \sum_{i=1}^{m}(r_i + 1)\ln(1 - x_{i:m:n}^{\alpha}).$$

Proof. See Appendix D. □

Denote the MLE of α derived from (8) in Theorem 4 as $\acute{\alpha}$, then from Theorem 3 the MLE of β_k, $k = 1, 2$, say $\acute{\beta}_k$, is given by

$$\acute{\beta}_k = -\frac{m_k}{\sum_{i=1}^{m}(r_i + 1)\ln(1 - x_{i:m:n}^{\acute{\alpha}})}, \quad k = 1, 2. \tag{9}$$

• Inference for common parameters $\beta_1 = \beta_2 = \beta$.

Theorem 5. *Suppose the latent competing risks failure times come from Kumaraswamy distribution with parameters* (α_1, β) *and* (α_2, β)*, for* $m > 0$ *and given* α_1 *and* α_2*, the MLE of* β *exists and is given by*

$$\dddot{\beta} = -\frac{m}{\sum_{k=1}^{2}\sum_{i=1}^{m}(r_i+1)\ln(1-x_{i:m:n}^{\alpha_k})}.$$

Proof. See Appendix E. □

Theorem 6. *Suppose the latent competing risks failure times come from Kumaraswamy distribution with parameters* (α_1, β) *and* (α_2, β)*, the MLEs of* α_1 *and* α_2 *can be obtained from following nonlinear equation*

$$\begin{cases} \dfrac{m_1}{\alpha_1} + m\dfrac{\sum_{i=1}^{m}(r_i+1)\frac{x_{i:m:n}^{\alpha_1}\ln x_{i:m:n}}{1-x_{i:m:n}^{\alpha_1}}}{\sum_{k=1}^{2}\sum_{i=1}^{m}(r_i+1)\ln(1-x_{i:m:n}^{\alpha_k})} + \sum_{i=1}^{m}\dfrac{\mathbb{1}(\delta_i=1)\ln x_{i:m:n}}{1-x_{i:m:n}^{\alpha_1}} = 0, \\ \dfrac{m_2}{\alpha_2} + m\dfrac{\sum_{i=1}^{m}(r_i+1)\frac{x_{i:m:n}^{\alpha_2}\ln x_{i:m:n}}{1-x_{i:m:n}^{\alpha_2}}}{\sum_{k=1}^{2}\sum_{i=1}^{m}(r_i+1)\ln(1-x_{i:m:n}^{\alpha_k})} + \sum_{i=1}^{m}\dfrac{\mathbb{1}(\delta_i=2)\ln x_{i:m:n}}{1-x_{i:m:n}^{\alpha_2}} = 0. \end{cases} \tag{10}$$

Proof. Substituting $\dddot{\beta}$ to the log-likelihood function $\ell(\alpha_1, \alpha_2, \beta)$ shown in Appendix E and taking derivative for it with respect to α_1 and α_2, the result can be obtained directly which is omitted. □

Let $\grave{\alpha}_1$ and $\grave{\alpha}_2$ be the MLE of α derived from (10) in Theorem 6, then from Theorem 5 the MLE of β, namely $\grave{\beta}$, exists and can be written as

$$\grave{\beta} = -\frac{m}{\sum_{k=1}^{2}\sum_{i=1}^{m}(r_i+1)\ln(1-x_{i:m:n}^{\grave{\alpha}_k})}. \tag{11}$$

3.3. Approximate confidence intervals

In this section, the associated approximate confidence intervals (ACIs) of unknown parameters are constructed by using the asymptotic distribution of MLEs.

Denote $\lambda = (\beta_1, \beta_2, \alpha_1, \alpha_2)'$ with $\lambda_k = \beta_k$, $k = 1, 2$ and $\lambda_k = \alpha_{k-2}$, $k = 3, 4$, the fisher information matrix is given by

$$I(\lambda) = \left[-\frac{\partial^2\ell(\beta_1, \beta_2, \alpha_1, \alpha_2)}{\partial\lambda_j\partial\lambda_k}\right]_{j,k=1,2,3,4} = \left[I_{jk}\right]_{4\times4},$$

where the elements of $I(\lambda)$ are provided in Appendix F.

Under some mild regularity, it is known the asymptotic distribution of the MLE $\hat{\lambda}$ is $\hat{\lambda} - \lambda \to N(0, I^{-1}(\hat{\lambda}))$, where '$\to$' means 'distributed as', and $I^{-1}(\hat{\lambda})$ is the inverse of the observed information matrix given by

$$I^{-1}(\hat{\lambda}) = \begin{pmatrix} Var(\hat{\beta}_1) & Cov(\hat{\beta}_1, \hat{\beta}_2) & Cov(\hat{\beta}_1, \hat{\alpha}_1) & Cov(\hat{\beta}_1, \hat{\alpha}_2) \\ Cov(\hat{\beta}_2, \hat{\beta}_1) & Var(\hat{\beta}_2) & Cov(\hat{\beta}_2, \hat{\alpha}_1) & Cov(\hat{\beta}_2, \hat{\alpha}_2) \\ Cov(\hat{\alpha}_1, \hat{\beta}_1) & Cov(\hat{\alpha}_1, \hat{\beta}_2) & Var(\hat{\alpha}_1) & Cov(\hat{\alpha}_1, \hat{\alpha}_2) \\ Cov(\hat{\alpha}_2, \hat{\beta}_1) & Cov(\hat{\alpha}_2, \hat{\alpha}_2) & Cov(\hat{\alpha}_2, \hat{\alpha}_1) & Var(\hat{\alpha}_2) \end{pmatrix}.$$

Hence , for arbitrary $0 < \gamma < 1$ the $100(1-\gamma)\%$ ACI of λ_j can be written as

$$\left(\hat{\lambda}_j - z_{\gamma/2}\sqrt{Var(\hat{\lambda}_j)}, \hat{\lambda}_j + z_{\gamma/2}\sqrt{Var(\hat{\lambda}_j)}\right), j = 1, 2, 3, 4,$$

where z_γ is the upper γth quantile of the standard normal distribution.

Sometimes, the ACIs obtained by previous procedure may have a negative lower bound. In order to overcome this drawback, the logarithmic transformation and delta methods can be used to obtain the asymptotic normality distribution of $\ln\hat{\lambda}_j$ as $\frac{\ln\hat{\lambda}_j - \ln\lambda_j}{Var(\ln\hat{\lambda}_j)} \to N(0, 1)$, $i = 1, 2, 3, 4$. Therefore, a $100(1-\gamma)\%$ normal-approximate confidence interval of λ_j derived in this manner can be constructed as

$$\left(\frac{\hat{\lambda}_j}{\exp\left(z_{\gamma/2}\sqrt{\widehat{Var}(\ln\hat{\lambda}_j)}\right)}, \hat{\lambda}_j\exp\left(z_{\gamma/2}\sqrt{\widehat{Var}(\ln\hat{\lambda}_j)}\right)\right),$$

where $\widehat{Var}(\ln\hat{\lambda}_j) = \widehat{Var}(\hat{\lambda}_j)/\hat{\lambda}_j, j = 1, 2, 3, 4$.

Moreover, when the latent failure times follow the Kumaraswamy competing risks model with common parameters $\alpha_1 = \alpha_2 = \alpha$ or $\beta_1 = \beta_2 = \beta$, the ACIs for parameters $(\beta_1, \beta_2, \alpha)$ or $(\alpha_1, \alpha_2, \beta)$ can be obtained similarly which are not reported here.

4. Bayesian inference

In recent decades, the Bayesian viewpoint, as a powerful and valid alternative to traditional frequentist perspective, has received widely attention in statistical inference. In this section, estimates including Bayes estimates and associated Bayes highest posterior density credible intervals (BCIs) for unknown parameters are obtained under squared error loss.

4.1. Prior information

As suggested by Nadar et al. [14], it is assumed that parameters β_1, β_2 and α_1, α_2 are independent random variables and are gamma distributed with PDFs respectively as

$$\pi_1(\beta_1) = \frac{b_1^{a_1}}{\Gamma(a_1)}\beta_1^{a_1-1}e^{-b_1\beta_1},\ a_1 > 0, b_1 > 0, \beta_1 > 0,$$
$$\pi_2(\beta_2) = \frac{b_2^{a_2}}{\Gamma(a_2)}\beta_2^{a_2-1}e^{-b_2\beta_2},\ a_2 > 0, b_2 > 0, \beta_2 > 0,$$
$$\pi_3(\alpha_1) = \frac{b_3^{a_3}}{\Gamma(a_3)}\alpha_1^{a_3-1}e^{-b_3\alpha_1},\ a_3 > 0, b_3 > 0, \alpha_1 > 0,$$
$$\pi_4(\alpha_2) = \frac{b_4^{a_4}}{\Gamma(a_4)}\alpha_2^{a_4-1}e^{-b_4\alpha_2},\ a_4 > 0, b_4 > 0, \alpha_2 > 0.$$

These prior distributions are also used for one-sample Kumaraswamy distribution by Ghosh and Nadarajah [15] under simple Type-II censored sample case. Moreover, the joint prior density of $(\beta_1, \beta_2, \alpha_1, \alpha_2)$, say $\pi(\beta_1, \beta_2, \alpha_1, \alpha_2)$, is given by

$$\pi(\beta_1, \beta_2, \alpha_1, \alpha_4) = \pi_1(\beta_1)\pi_2(\beta_2)\pi_3(\alpha_1)\pi_4(\alpha_2), \tag{12}$$

and the joint posterior PDF of β_1, β_2 and α_1, α_2, namely $\pi(\beta_1, \beta_2, \alpha|x)$, can be obtained from (12) as

$$\pi(\beta_1, \beta_2, \alpha_1, \alpha_2|x) = \frac{\pi(\beta_1, \beta_2, \alpha_1, \alpha_2)L(\beta_1, \beta_2, \alpha_1, \alpha_2; x)}{\int_{(4)} \pi(\beta_1, \beta_2, \alpha_1, \alpha_2)L(\beta_1, \beta_2, \alpha_1, \alpha_2; x)d\lambda}, \tag{13}$$

where notation $\int_{(4)}(\cdot)d\lambda = \int_0^\infty \int_0^\infty \int_0^\infty \int_0^\infty (\cdot)d\beta_1 d\beta_2 d\alpha_1 d\alpha_2$. Therefore, under squared error loss, the Bayes estimator of any function of β_1, β_2 and α_1, α_2, say $\eta(\beta_1, \beta_2, \alpha_1, \alpha_2)$, is the mean of the posterior distribution as

$$\hat{\eta}(\beta_1, \beta_2, \alpha_1, \alpha_2) = \frac{\int_{(4)} \eta(\beta_1, \beta_2, \alpha_1, \alpha_2)\pi(\beta_1, \beta_2, \alpha_1, \alpha_2)L(\beta_1, \beta_2, \alpha_1, \alpha_2; x)d\lambda}{\int_{(4)} \pi(\beta_1, \beta_2, \alpha_1, \alpha_2)L(\beta_1, \beta_2, \alpha_1, \alpha_2; x)d\lambda}.$$

It is clear that the corresponding Bayes estimator of $\eta(\beta_1, \beta_2, \alpha_1, \alpha_2)$ is expressed as the ratio of two integrals which cannot be expressed in closed form, thus Monte-Carlo Markov chain (MCMC) sampling method is used to approximate the associated Bayes estimate.

4.2. Posterior analysis and MCMC method

Using (4) and (12) and ignoring the additive constant, the joint posterior PDF (13) of β_1, β_2 and α_1, α_2 can be rewritten as

$$\begin{aligned}\pi(\beta_1, \beta_2, \alpha_1, \alpha_2|x) \propto\ & \beta_1^{a_1+m_1-1} \exp\left\{-\beta_1\left[b_1 - \sum_{i=1}^{m}(r_i+1)\ln(1-x_{i:m:n}^{\alpha_1})\right]\right\} \\ & \cdot\ \beta_2^{a_2+m_2-1} \exp\left\{-\beta_2\left[b_2 - \sum_{i=1}^{m}(r_1+1)\ln(1-x_{i:m:n}^{\alpha_2})\right]\right\} \\ & \cdot\ \alpha_1^{a_3+m_1-1} \exp\left\{-\alpha_1\left[b_3 - \sum_{i=1}^{m}\mathbb{1}(\delta_i=1)\ln x_{i:m:n}\right]\right\} \prod_{i=1}^{m}\frac{1}{[1-x_{i:m:n}^{\alpha_1}]^{\mathbb{1}(\delta_i=1)}} \\ & \cdot\ \alpha_2^{a_4+m_2-1} \exp\left\{-\alpha_2\left[b_4 - \sum_{i=1}^{m}\mathbb{1}(\delta_i=2)\ln x_{i:m:n}\right]\right\} \prod_{i=1}^{m}\frac{1}{[1-x_{i:m:n}^{\alpha_2}]^{\mathbb{1}(\delta_i=2)}}.\end{aligned} \tag{14}$$

From (14), it is seen that the posterior PDF of β_k, $k = 1, 2$ given α_k is gamma distributed, of which the joint conditional posterior PDF of β_1 and β_2 given α_1 and α_2, namely $\pi(\beta_1, \beta_2|\alpha_1, \alpha_2, x)$, is given by

$$\pi(\beta_1, \beta_2|\alpha_1, \alpha_2, x) = \pi(\beta_1|\alpha_1, x)\pi(\beta_2|\alpha_2, x), \tag{15}$$

with

$$\pi(\beta_k|\alpha_k, x) \propto \beta_k^{a_k+m_k-1} \exp\left\{-\beta_k\left[b_k - \sum_{i=1}^{m}(r_i+1)\ln(1-x_{i:m:n}^{\alpha_k})\right]\right\}, \ k = 1, 2.$$

Meanwhile, the conditional posterior PDF of α_1 and α_2, say $\pi(\alpha_1, \alpha_2|\beta_1, \beta_2, x)$, is

$$\pi(\alpha_1, \alpha_2|\beta_1, \beta_2, x) = \pi(\alpha_1|\beta_1, x)\pi(\alpha_2|\beta_2, x), \tag{16}$$

with

$$\pi(\alpha_k|\beta_k, x) \propto \alpha_k^{a_{k+1}+m_k-1} \exp\left\{-\alpha_k\left[b_{k+1} - \sum_{i=1}^{m}\mathbb{1}(\delta_i = k)\ln x_{i:m:n}\right]\right\}$$
$$\cdot \prod_{i=1}^{m}(1-x_{i:m:n}^{\alpha_k})^{(r_i+1)\beta_k - \mathbb{1}(\delta_i=k)}, \qquad k = 1, 2.$$

From $\partial^2 \ln \pi(\alpha_k|\beta_k, x)/\partial\alpha_k^2$, it is noted by direct computation that function $\pi(\alpha_k|\beta_k, x)$ is log-concave for $\beta_k \geq 1$, and that one can use the general method of Devroye [16] to generate random sample from the log-concave density $\pi(\alpha_k|\beta_k, x)$. Meanwhile, for $\beta_k < 1$, $\pi(\alpha_k|\beta_k, x)$ can be rewritten as

$$\pi(\alpha_k|\beta_k, x) \propto e^{\alpha_k \ln x_{1:m:n}}(1-e^{\alpha_k \ln x_{1:m:n}})^{(r_i+1)\beta_k - 1}\Bigg[(1-x_{1:m:n}^{\alpha_k})^{1-\mathbb{1}(\delta_i=k)}$$
$$\cdot \alpha_k^{a_{k+1}+m_k-1} e^{-\alpha_k\left[\ln x_{1:m:n} + b_{k+1} - \sum_{i=1}^{m}\mathbb{1}(\delta_i=k)\ln x_{i:m:n}\right]} \prod_{i=2}^{m}(1-x_{i:m:n}^{\alpha_k})^{(r_i+1)\beta_k - \mathbb{1}(\delta_i=k)}\Bigg].$$

Observe that the continuous function within $[\cdot]$ goes to 0 as α_k approaches 0 and goes to a finite constant as α_k approaches ∞, which yields the function has a finite maximum. Therefore, random sample of β_k with respect to $\beta_k < 1$ can be generated by using the acceptance–rejection principle based on the generalized exponential generator.

Following MCMC procedure is provided to generate random sample and compute associated Bayes estimates.

MCMC sampling algorithm

Step 1 Use the method of MLEs or any other methods to estimate the parameters β_1, β_2 and α_1, α_2, as starting point of iteration, denote these estimates as $\beta_1^{(0)}$, $\beta_2^{(0)}$ and $\alpha_1^{(0)}$, $\alpha_2^{(0)}$.

Step 2 In kth iteration, let $\beta_1^{(k)}$, $\beta_2^{(k)}$ and $\alpha_1^{(k)}$, $\alpha_2^{(k)}$ be the estimates of β_1, β_2 and α_1, α_2, then we generate $\beta_1^{(k+1)}$, $\beta_2^{(k+1)}$ and $\alpha_1^{(k+1)}$, $\alpha_2^{(k+1)}$ from densities $\pi(\beta_1, \beta_2|\alpha_1^{(k)}, \alpha_2^{(k)}, x)$ and $\pi(\alpha_1, \alpha_2|\beta_1^{(k)}, \beta_2^{(k)}, x)$, respectively.

Step 3 Repeat Step 2, K times.

Step 4 Compute Bayes estimate of $\eta(\beta_1, \beta_2, \alpha_1, \alpha_2)$ by

$$\hat{\eta}(\beta_1, \beta_2, \alpha_1, \alpha_2) = \frac{1}{K-K_0}\sum_{k=K_0+1}^{K}\eta(\beta_1^{(k)}, \beta_2^{(k)}, \alpha_1^{(k)}, \alpha_2^{(k)}),$$

where K_0 is burn-in period.

Step 5 To construct the BCI of $\eta(\beta_1, \beta_2, \alpha_1, \alpha_2)$, arrange all estimates $\hat{\eta}_k = \hat{\eta}(\beta_1^{(k)}, \beta_2^{(k)}, \alpha_1^{(k)}, \alpha_2^{(k)})$, $k = K_0+1, K_0+2, \ldots, K$ in an ascend order, as $\hat{\eta}^{[1]}, \hat{\eta}^{[2]}, \ldots, \hat{\eta}^{[K-K_0]}$, then for arbitrary $0 < \gamma < 1$, a $100(1-\gamma)$ credible interval of $\eta(\beta_1, \beta_2, \alpha_1, \alpha_2)$ can be obtained as

$$(\eta^{[l]}, \eta^{[l+K-K_0-[\gamma(K-K_0)+1]]}), \ l = 1, 2, \ldots, [(K-K_0)\gamma]$$

where $[y]$ denotes the greatest integer less than or equal to y. Therefore, the $100(1-\gamma)\%$ high posterior density BCI can be constructed as the l^*th one satisfying

$$\eta^{[l^*+K-K_0-[\gamma(K-K_0)+1]]} - \eta^{[l^*]} = \min_{l=1}^{(K-K_0)\gamma}(\eta^{[l+K-K_0-[\gamma(K-K_0)+1]]} - \eta^{[l]}).$$

Note that when $a_1 = b_1 = a_2 = b_2 = a_3 = b_3 = a_4 = b_4 = 0$, the prior (12) is not proper, but the conditional posterior densities $\pi(\beta_1, \beta_2|\alpha_1, \alpha_2, x)$ and $\pi(\alpha_1, \alpha_2|\beta_1, \beta_2, x)$ are still proper. Hence, previous sampling method can be still used to find corresponding Bayes estimates.

4.3. Bayesian inference with common parameters

In this subsection, analysis will be presented under Bayesian perspective when there are common Kumaraswamy parameters.

• Inference for common parameters $\alpha_1 = \alpha_2 = \alpha$.

When parameters α_1 and α_2 are equal, independent gamma priors for parameters β_1, β_2 and α are suggested as follows

$$\pi_1(\beta_1) = \frac{b_1^{a_1}}{\Gamma(a_1)}\beta_1^{a_1-1}e^{-b_1\beta_1},\ a_1 > 0, b_1 > 0, \beta_1 > 0,$$
$$\pi_2(\beta_2) = \frac{b_2^{a_2}}{\Gamma(a_2)}\beta_2^{a_2-1}e^{-b_2\beta_2},\ a_2 > 0, b_2 > 0, \beta_2 > 0,$$
$$\pi_3(\alpha) = \frac{b_3^{a_3}}{\Gamma(a_3)}\alpha^{a_3-1}e^{-b_3\alpha},\ a_3 > 0, b_3 > 0, \alpha > 0.$$

Following the similar procedure as previous, the joint posterior PDF of β_1 and β_2 and α can be expressed as

$$\begin{aligned}\pi(\beta_1, \beta_2, \alpha|x) \propto\ & \beta_1^{m_1+a_1-1}\exp\left\{-\beta_1[b_1 - \sum_{i=1}^{m}(r_i+1)\ln(1-x_{i:m:n}^{\alpha})]\right\}\\ & \cdot\ \beta_2^{m_2+a_2-1}\exp\left\{-\beta_2[b_2 - \sum_{i=1}^{m}(r_i+1)\ln(1-x_{i:m:n}^{\alpha})]\right\}\\ & \cdot\ \alpha^{m+a_3-1}\exp\left\{-\alpha[b_3 - \sum_{i=1}^{m}\ln x_{i:m:n}]\right\}\prod_{i=1}^{m}\frac{1}{1-x_{i:m:n}^{\alpha}}.\end{aligned} \tag{17}$$

From (17), the joint conditional posterior PDF of β_1 and β_2 given α, namely $\pi(\beta_1, \beta_2|\alpha, x)$, is given by

$$\begin{aligned}\pi(\beta_1, \beta_2|\alpha, x) \propto\ & \beta_1^{m_1+a_1-1}\exp\left\{-\beta_1[b_1 - \sum_{i=1}^{m}(r_i+1)\ln(1-x_i^{\alpha})]\right\}\\ & \cdot\ \beta_2^{m_2+a_2-1}\exp\left\{-b_2[\beta_2 - \sum_{i=1}^{m}(r_i+1)\ln(1-x_i^{\alpha})]\right\},\end{aligned} \tag{18}$$

and the associated conditional posterior PDF of α given β_1 and β_2, say $\pi(\alpha|\beta_1, \beta_2, x)$, is given by

$$\begin{aligned}\pi(\alpha|\beta_1, \beta_2, x) &\propto \alpha^{m+a_3-1}e^{-\alpha[b_3-\sum_{i=1}^{m}\ln x_{i:m:n}]}\prod_{i=1}^{m}(1-x_{i:m:n}^{\alpha})^{(r_i+1)(\beta_1+\beta_2)-1},\\ &\propto (1-e^{\alpha\ln x_{1:m:n}})^{(r_i+1)(\beta_1+\beta_2)-1}e^{\alpha\ln x_{1:m:n}}\Bigg[\alpha^{m+a_3-1}\\ &\cdot\ \exp\left\{-\alpha[b_3 - \sum_{i=2}^{m}\ln x_{i:m:n}]\right\}\prod_{i=2}^{m}(1-x_{i:m:n}^{\alpha})^{(r_i+1)(\beta_1+\beta_2)-1}\Bigg].\end{aligned} \tag{19}$$

It is also observed that $\pi(\alpha|\beta_1, \beta_2, x)$ is log-concave for $\beta_1 + \beta_2 \geq 1$, and has a finite maximum when $\beta_1 + \beta_2 < 1$.

• Inference for common parameters $\beta_1 = \beta_2 = \beta$.

When parameters β_1 and β_2 are equal, independent gamma priors for parameters α_1, α_2 and β are suggested as follows

$$\pi_1(\beta) = \frac{b_1^{a_1}}{\Gamma(a_1)}\beta^{a_1-1}e^{-b_1\beta},\ a_1 > 0, b_1 > 0, \beta > 0,$$
$$\pi_3(\alpha_1) = \frac{b_3^{a_3}}{\Gamma(a_3)}\alpha_1^{a_3-1}e^{-b_3\alpha_1},\ a_3 > 0, b_3 > 0, \alpha_1 > 0,$$
$$\pi_4(\alpha_2) = \frac{b_4^{a_4}}{\Gamma(a_4)}\alpha_2^{a_4-1}e^{-b_4\alpha_2},\ a_4 > 0, b_4 > 0, \alpha_2 > 0.$$

Therefore, the joint posterior PDF of α_1, α_2 and β is

$$\begin{aligned}\pi(\beta, \alpha_1, \alpha_2|x) \propto\ & \beta^{a_1+m-1}\exp\left\{-\beta\left[b_1 - \sum_{i=1}^{m}(r_i+1)\ln[(1-x_{i:m:n}^{\alpha_1})(1-x_{i:m:n}^{\alpha_2})]\right]\right\}\\ & \cdot\ \alpha_1^{a_3+m_1-1}\exp\left\{-\alpha_1\left[b_3 - \sum_{i=1}^{m}\mathbb{1}(\delta_i=1)\ln x_{i:m:n}\right]\right\}\cdot\prod_{i=1}^{m}[1-x_{i:m:n}^{\alpha_1}]^{-\mathbb{1}(\delta_i=1)}\end{aligned}$$

$$\cdot\ \alpha_2^{a_4+m_2-1}\exp\left\{-\alpha_2\left[b_4-\sum_{i=1}^{m}\mathbb{1}(\delta_i=2)\ln x_{i:m:n}\right]\right\}\cdot\prod_{i=1}^{m}[1-x_{i:m:n}^{\alpha_2}]^{-\mathbb{1}(\delta_i=2)}. \tag{20}$$

From (20), one further has

$$\pi(\beta|\alpha_1,\alpha_2,x)\propto\beta^{a_1+m-1}\exp\left\{-\beta\left[b_1-\sum_{i=1}^{m}(r_i+1)\ln[(1-x_{i:m:n}^{\alpha_1})(1-x_{i:m:n}^{\alpha_2})]\right]\right\}, \tag{21}$$

and

$$\pi(\alpha_1,\alpha_2|\beta,x)=\pi(\alpha_1|\beta,x)\pi(\alpha_1|\beta,x), \tag{22}$$

with

$$\begin{aligned}\pi(\alpha_k|\beta,x)&\propto\alpha_k^{a_{k+2}+m_k-1}\exp\left\{-\alpha_k\left[b_{k+2}-\sum_{i=1}^{m}\mathbb{1}(\delta_i=k)\ln x_{i:m:n}\right]\right\}\\&\quad\cdot\prod_{i=1}^{m}[1-x_{i:m:n}^{\alpha_k}]^{(r_i+1)\beta-\mathbb{1}(\delta_i=k)},\\&\propto[1-x_{1:m:n}^{\alpha_k}]^{(r_i+1)\beta-1}x_{1:m:n}^{\alpha_k}\Big[(1-x_{1:m:n}^{\alpha_k})^{1-\mathbb{1}(\delta_i=k)}\alpha_k^{a_{k+2}+m_k-1}\\&\quad\cdot\exp\left\{-\alpha_k\left[\ln x_{1:m:n}+b_{k+2}-\sum_{i=1}^{m}\mathbb{1}(\delta_i=k)\ln x_{i:m:n}\right]\right\}\\&\quad\cdot\prod_{i=2}^{m}[1-x_{i:m:n}^{\alpha_k}]^{(r_i+1)\beta-\mathbb{1}(\delta_i=k)}\Big],\qquad k=1,2.\end{aligned}$$

One can observe that $\pi(\alpha_k|\beta,x)$ is log-concave for $\beta\geq 1$, and has a finite maximum when $\beta<1$.

Therefore, when both cases of common Kumaraswamy parameters are available (i.e., $\alpha_1=\alpha_2=\alpha$ and $\beta_1=\beta_2=\beta$), the Bayesian point and credible interval estimates can be obtained by using similar algorithm as shown in Subsection 4.2 based on the method of Devroye [16] and the acceptance–rejection principle.

5. Some testing problems

As complementary and application, the equivalence of the parameters α_1,α_2 and β_1,β_2 among two different competing risks from the Kumaraswamy model is investigated under progressive censoring. Following hypothesis testings are proposed as

Test I $H_0:\alpha_1=\alpha_2=\alpha$ versus $H_1:\alpha_1\neq\alpha_2$.
Test II $H_0:\beta_1=\beta_2=\beta$ versus $H_1:\beta_1\neq\beta_2$.

In Test I, under H_0 and based on Theorems 3 and 4, the MLE of (β_1,β_2,α) is $(\acute{\beta}_1,\acute{\beta}_2,\grave{\alpha})$ and the associated log-likelihood function is $\ell(\beta_1,\beta_2,\alpha)$ shown in Appendix C. Therefore, for large n, the likelihood ratio statistic

$$-2\{\ell(\acute{\beta}_1,\acute{\beta}_2,\grave{\alpha})-\ell(\hat{\beta}_1,\hat{\beta}_2,\hat{\alpha}_1,\hat{\alpha}_2)\}\longrightarrow\chi_1^2, \tag{23}$$

where $\ell(\beta_1,\beta_2,\alpha_1,\alpha_2)$ is the log-likelihood function at (5) when parameters are different. Therefore, the asymptotic distribution of $-2\{\ell(\acute{\beta}_1,\acute{\beta}_2,\grave{\alpha})-\ell(\hat{\beta}_1,\hat{\beta}_2,\hat{\alpha}_1,\hat{\alpha}_2)\}$ provided in (23) can be used to construct the likelihood ratio test for Test I, and reject H_0 under this case if $-2\{\ell(\acute{\beta}_1,\acute{\beta}_2,\grave{\alpha})-\ell(\hat{\beta}_1,\hat{\beta}_2,\hat{\alpha}_1,\hat{\alpha}_2)\}>c^*$ where c^* is such that $P(\chi_1^2>c^*)=$ size of the test.

Meanwhile, in Test II, under H_0, the MLE of $(\alpha_1,\alpha_2,\beta)$ is obtained from Theorems 5 and 6 as $(\grave{\alpha}_1,\grave{\alpha}_2,\grave{\beta})$. Hence, for large n, the likelihood ratio statistic

$$-2\{\ell(\grave{\alpha}_1,\grave{\alpha}_2,\grave{\beta})-\ell(\hat{\beta}_1,\hat{\beta}_2,\hat{\alpha}_1,\hat{\alpha}_2)\}\sim\chi_1^2, \tag{24}$$

and the likelihood ratio test can be also constructed from (24) similarly as shown previously in Test I.

6. Numerical illustration

6.1. Simulation studies

In this subsection, simulations are provided to investigate the proposed methods. The performance of the associated results can be evaluated in terms of criteria quantities mean square error (MSE) and average bias (AB) for point estimates, and coverage probability (CP) and average width (AW) for interval estimates, and following different CSs are considered

Table 1
ABs and MSEs (within bracket) for model parameters with $(\beta_1, \beta_2, \alpha_1, \alpha_2) = (0.8, 0.9, 1.4, 1.2)$.

n	m	CS	MLE				Bayes			
			β_1	β_2	α_1	α_2	β_1	β_2	α_1	α_2
30	15	I	0.4934	0.5719	0.3892	0.4536	0.4652	0.5364	0.3652	0.4358
			[0.2835]	[0.3504]	[0.1622]	[0.2334]	[0.2451]	[0.3080]	[0.1565]	[0.2153]
		II	0.5014	0.5538	0.4157	0.4720	0.4878	0.5411	0.3470	0.4126
			[0.2718]	[0.3433]	[0.1938]	[0.2483]	[0.2549]	[0.3198]	[0.1643]	[0.1904]
		III	0.5102	0.5644	0.3948	0.4695	0.4711	0.5397	0.3548	0.4340
			[0.2832]	[0.3583]	[0.1714]	[0.2352]	[0.2422]	[0.3081]	[0.1404]	[0.2151]
	25	I	0.4369	0.4923	0.3149	0.3849	0.3910	0.4582	0.2756	0.3527
			[0.1983]	[0.2662]	[0.1116]	[0.1669]	[0.1774]	[0.2397]	[0.0948]	[0.1441]
		II	0.4128	0.4791	0.2906	0.3622	0.4024	0.4369	0.2841	0.3611
			[0.1888]	[0.2452]	[0.0982]	[0.1587]	[0.1699]	[0.2106]	[0.0846]	[0.1532]
		III	0.4415	0.4852	0.3083	0.3758	0.3956	0.4473	0.2799	0.3598
			[0.2064]	[0.2550]	[0.1098]	[0.1475]	[0.1783]	[0.2162]	[0.0924]	[0.1443]
60	30	I	0.2417	0.3568	0.1912	0.2413	0.2148	0.3483	0.1842	0.2261
			[0.0648]	[0.1407]	[0.0499]	[0.0751]	[0.0521]	[0.1330]	[0.0415]	[0.0635]
		II	0.2256	0.3812	0.2049	0.2665	0.1991	0.3549	0.1795	0.2148
			[0.0588]	[0.1562]	[0.0563]	[0.0826]	[0.0569]	[0.1472]	[0.0414]	[0.0553]
		III	0.2384	0.3610	0.2113	0.2619	0.2025	0.3396	0.2020	0.2309
			[0.0623]	[0.1517]	[0.0594]	[0.0858]	[0.0492]	[0.1284]	[0.0500]	[0.0587]
	45	I	0.1539	0.2439	0.1231	0.1994	0.1459	0.2260	0.1154	0.1643
			[0.0321]	[0.0696]	[0.0203]	[0.0497]	[0.0307]	[0.0632]	[0.0195]	[0.0384]
		II	0.1664	0.2661	0.1354	0.2012	0.1362	0.2195	0.1032	0.1582
			[0.0349]	[0.0820]	[0.0251]	[0.0451]	[0.0231]	[0.0553]	[0.0153]	[0.0359]
		III	0.1719	0.2512	0.1320	0.1843	0.1551	0.2304	0.1149	0.1705
			[0.0383]	[0.0776]	[0.0213]	[0.0535]	[0.0360]	[0.0631]	[0.0203]	[0.0384]

CS I: $r_1 = r_2 = \cdots = r_{m-1} = 0$ and $r_m = n - m$;
CS II: $r_1 = n - m$ and $r_2 = \cdots = r_m = 0$;
CS III: $\begin{cases} r_1 = \cdots = r_{n-m} = 1, r_{n-m+1} = \cdots = r_m = 0, & \text{if } n \le 2m; \\ r_1 = \cdots = r_{m-1} = 1, r_m = n - 2m + 1, & \text{if } n > 2m. \end{cases}$

Meanwhile, for generating the Kumaraswamy competing risks samples, progressively Type-II censored data are generated for each cause of failure by using the algorithm of Balakrishnan and Sandhu [17], and then the progressively competing risks data and their failure cause are obtained by comparison.

The criteria quantities for point and interval estimates were examined under different choice of $\beta_1, \beta_2, \alpha_1, \alpha_2, m, n$ and $r_1, r_2, \ldots, r_m$ values, and the simulation was repeated by 10,000 times following a burn-in phase of 1500 iterations. The results are presented in Tables 1 to 6, where the significance level is taken to be $\gamma = 0.05$ for interval estimates, and the Bayes estimates are evaluated with respect to the priors with all hyper-parameters being 0.001 (instead of 0.0) which are proper but almost non-informative.

From Tables 1, 3 and 5, one can observe for the point estimates that

- with the increase of n or m or both n and m, ABs and MSEs of all MLEs and Bayes estimates decrease;
- for fixed sample size n and m, ABs and MSEs of MLEs are similar under three CSs I, II and III, same phenomenon also appears for Bayes estimates, respectively;
- ABs and MSEs of Bayes estimates for unknown parameters are relatively smaller than that of MLEs.

Meanwhile, it is also seen from Tables 2, 4 and 6 for interval estimates that

- CPs increase when n, or m or both n and m increase, and the AWs decrease correspondingly under same change;
- for fixed n and m, ACIs have similar CPs and AWs under CSs I, II and III, this phenomenon also appear for BCIs.
- the ACIs and BCIs have similar CPs under same sample sizes and CSs which are close to the nominal significance level in most cases, whereas the AWs of BCIs have slight shorter AWs than that of associated ACIs.

To sum up, the performance of MLEs and Bayes estimates are satisfactory, whereas although the Bayes estimates are obtained under almost non-informative prior, they are slightly superior to MLEs in terms of ABs and MSEs. Meanwhile, according to criteria quantities AWs and CPs, if one want to find intervals featuring shorter AWs, BCIs may be an appropriate choice; otherwise if one wants to have intervals of which the CP close to the nominal level and the AW is not a major concern, the ACIs can be used which provide a balance between CP and AW.

6.2. Real-life example

In this section, we present the analysis of a real-life data set for illustrative purpose. The data under consideration correspond to the football data where at least one goal scored by the home team and at least one goal scored directly from

Table 2
CPs and AWs (within bracket) for model parameters with $(\beta_1, \beta_2, \alpha_1, \alpha_2) = (0.8, 0.9, 1.4, 1.2)$.

n	m	CS	ACI				BCI			
			β_1	β_2	α_1	α_2	β_1	β_2	α_1	α_2
30	15	I	0.9226	0.9251	0.9183	0.9210	0.9207	0.9247	0.9177	0.9204
			[1.2129]	[1.3238]	[0.8435]	[0.9721]	[0.9523]	[1.1219]	[0.7624]	[0.9231]
		II	0.9215	0.9263	0.9201	0.9206	0.9198	0.9269	0.9159	0.9193
			[0.9938]	[1.4012]	[0.8312]	[0.9538]	[0.9412]	[1.2537]	[0.7359]	[0.9068]
		III	0.9210	0.9249	0.9194	0.9198	0.9205	0.9258	0.9168	0.9210
			[1.1345]	[1.3629]	[0.8506]	[0.9616]	[0.9631]	[1.1963]	[0.7596]	[0.9204]
	25	I	0.9348	0.9359	0.9256	0.9294	0.9335	0.9341	0.9242	0.9278
			[0.9371]	[1.1226]	[0.6648]	[0.7859]	[0.8429]	[0.9438]	[0.5937]	[0.7549]
		II	0.9329	0.9376	0.9285	0.9290	0.9240	0.9329	0.9268	0.9261
			[0.9489]	[1.2045]	[0.6439]	[0.7660]	[0.8567]	[0.9256]	[0.6124]	[0.7428]
		III	0.9338	0.9348	0.9271	0.9302	0.9332	0.9336	0.9254	0.9283
			[0.9652]	[0.9941]	[0.6527]	[0.7913]	[0.8490]	[0.9362]	[0.6065]	[0.9493]
60	30	I	0.9417	0.9421	0.9344	0.9365	0.9396	0.9412	0.9326	0.9355
			[0.6135]	[0.7703]	[0.4352]	[0.5442]	[0.4619]	[0.7244]	[0.4023]	[0.4812]
		II	0.9409	0.9430	0.9362	0.9353	0.9410	0.9424	0.9321	0.9460
			[0.5826]	[0.7534]	[0.4179]	[0.5215]	[0.4852]	[0.7351]	[0.3852]	[0.4930]
		III	0.9396	0.9415	0.9379	0.9382	0.9407	0.9427	0.9340	0.9354
			[0.6038]	[0.7817]	[0.4311]	[0.5283]	[0.4718]	[0.7506]	[0.3961]	[0.4879]
	45	I	0.9463	0.9490	0.9428	0.9441	0.9449	0.9453	0.9419	0.9427
			[0.3524]	[0.2832]	[0.3168]	[0.4232]	[0.3334]	[0.2542]	[0.2627]	[0.3827]
		II	0.9471	0.9478	0.9419	0.9452	0.9438	0.9449	0.9405	0.9420
			[0.3819]	[0.2791]	[0.3048]	[0.4417]	[0.3160]	[0.2369]	[0.2453]	[0.3659]
		III	0.9459	0.9464	0.9416	0.9447	0.9442	0.9460	0.9413	0.9436
			[0.3605]	[0.2918]	[0.2919]	[0.4290]	[0.3258]	[0.2434]	[0.2490]	[0.3905]

Table 3
ABs and MSEs (within bracket) for model parameters with $(\beta_1, \beta_2, \alpha) = (1.5, 2.4, 0.8)$.

n	m	CS	MLE			Bayes		
			β_1	β_2	α	β_1	β_2	α
30	15	I	1.1272[1.3136]	1.1664[1.1872]	0.1690[0.1171]	0.9846[1.1424]	1.1103[1.0275]	0.1702[0.1213]
		II	1.1121[1.2998]	1.0743[1.1326]	0.1873[0.1199]	1.0672[0.9867]	1.1057[1.1000]	0.1658[0.1159]
		III	1.1063[1.3211]	1.1329[1.1689]	0.1745[0.1204]	1.0545[1.1293]	0.9879[1.0366]	0.1645[0.1162]
	25	I	0.8329[0.7108]	0.9279[1.0236]	0.1576[0.0847]	0.7981[0.6543]	0.9142[0.9752]	0.1423[0.0798]
		II	0.8457[0.6524]	0.9468[1.0517]	0.1496[0.0765]	0.8213[0.6601]	0.9084[0.9946]	0.1398[0.0642]
		III	0.8399[0.6899]	0.9289[1.0364]	0.1528[0.0912]	0.8242[0.6496]	0.9177[1.0101]	0.1466[0.0735]
60	30	I	0.5469[0.3089]	0.6984[0.5762]	0.1362[0.0360]	0.5068[0.2549]	0.6549[0.5324]	0.1247[0.0225]
		II	0.6143[0.4299]	0.7126[0.5664]	0.1328[0.0413]	0.4892[0.2891]	0.6873[0.5478]	0.1356[0.0346]
		III	0.5064[0.3166]	0.7248[0.6147]	0.1292[0.0451]	0.5110[0.2768]	0.6981[0.5812]	0.1259[0.0368]
	45	I	0.3009[0.1149]	0.6437[0.5100]	0.1097[0.0201]	0.2845[0.0969]	0.6247[0.4365]	0.0975[0.0112]
		II	0.2942[0.0976]	0.6594[0.4956]	0.1103[0.0196]	0.2953[0.1010]	0.6385[0.4297]	0.1011[0.0147]
		III	0.2836[0.0987]	0.6711[0.5232]	0.0998[0.0121]	0.2736[0.0955]	0.6291[0.4143]	0.1004[0.0123]

Table 4
CPs and AWs (within bracket) for model parameters with $(\beta_1, \beta_2, \alpha) = (1.5, 2.4, 0.8)$.

n	m	CS	ACI			BCI		
			β_2	β_1	α	β_1	β_2	α
30	15	I	0.9167[3.0415]	0.9258[4.1695]	0.9216[0.4466]	0.9103[2.5783]	0.9147[3.6849]	0.9148[0.4231]
		II	0.9113[2.7823]	0.9177[3.9649]	0.9148[0.4124]	0.9111[2.7451]	0.9134[3.5467]	0.9125[0.4079]
		III	0.9124[2.9546]	0.9235[4.1211]	0.9196[0.4237]	0.9108[2.6550]	0.9142[3.5843]	0.9098[0.3985]
	25	I	0.9231[2.2218]	0.9342[3.5497]	0.9289[0.3948]	0.9169[1.8970]	0.9276[3.2115]	0.9243[0.3658]
		II	0.9219[2.1004]	0.9329[3.5123]	0.9310[0.4105]	0.9181[1.9236]	0.9234[3.1111]	0.9218[0.3441]
		III	0.9206[2.0617]	0.9316[3.4986]	0.9304[0.4069]	0.9200[1.9928]	0.9289[3.2342]	0.9226[0.3529]
60	30	I	0.9378[1.6829]	0.9410[2.9159]	0.9346[0.2899]	0.9312[1.4396]	0.9356[2.7548]	0.9312[0.2566]
		II	0.9336[1.6748]	0.9435[3.0672]	0.9365[0.2973]	0.9295[1.3807]	0.9411[2.8005]	0.9299[0.2488]
		III	0.9410[1.8193]	0.9427[2.9887]	0.9382[0.3001]	0.9301[1.4112]	0.9384[2.6832]	0.9324[0.2654]
	45	I	0.9510[0.9121]	0.9523[1.1986]	0.9465[0.1934]	0.9446[0.7682]	0.9463[0.9897]	0.9433[0.1979]
		II	0.9484[0.8978]	0.9511[1.2052]	0.9478[0.2145]	0.9479[0.8145]	0.9488[1.0101]	0.9478[0.2033]
		III	0.9502[0.9065]	0.9498[1.1894]	0.9502[0.2314]	0.9453[0.8023]	0.9471[0.9964]	0.9500[0.2124]

a penalty kick, the origin data set was reported by Meintanis [18]. In this data analysis, random variable $X_{1\cdot}$ denotes the time in minutes of the first kick goal scored by any team and $X_{2\cdot}$ represents the first goal of any type scored by the home team, and the detail data is provided in Table 7, where each data points have been divided by 90 to guarantee that the data belong to the interval (0, 1), that is to say, the data can be viewed as the proportion of time that any team and the home

Table 5
ABs and MSEs (within bracket) for model parameters with $(\alpha_1, \alpha_2, \beta) = (2.3, 1.9, 1)$.

n	m	CS	MLE			Bayes		
			α_1	α_2	β	α_1	α_2	β
30	15	I	0.3231[0.1202]	0.3746[0.1544]	0.4453[0.2271]	0.3029[0.1134]	0.3443[0.1354]	0.3859[0.1731]
		II	0.3426[0.1419]	0.3652[0.1478]	0.4237[0.2169]	0.2954[0.0985]	0.3507[0.1385]	0.3940[0.1849]
		III	0.3390[0.1252]	0.3721[0.1509]	0.4296[0.2087]	0.3112[0.1073]	0.3486[0.1367]	0.4013[0.1727]
	25	I	0.2574[0.0748]	0.2895[0.0954]	0.3563[0.1419]	0.2243[0.0641]	0.2759[0.0826]	0.3215[0.1269]
		II	0.2459[0.0720]	0.2914[0.1044]	0.3447[0.1351]	0.2308[0.0704]	0.2543[0.0913]	0.3324[0.1195]
		III	0.2691[0.0857]	0.2945[0.1019]	0.3319[0.1412]	0.2279[0.0672]	0.2628[0.0887]	0.3146[0.1146]
60	30	I	0.1658[0.0372]	0.1848[0.0455]	0.2447[0.0706]	0.1436[0.0282]	0.1640[0.0400]	0.2218[0.0620]
		II	0.1563[0.0357]	0.2012[0.0560]	0.2251[0.0662]	0.1340[0.0311]	0.1752[0.0389]	0.2069[0.0671]
		III	0.1596[0.0320]	0.1937[0.0457]	0.2406[0.0740]	0.1297[0.0255]	0.1698[0.0412]	0.2097[0.0556]
	45	I	0.1211[0.0252]	0.1456[0.0321]	0.1723[0.0389]	0.0895[0.0134]	0.1211[0.0245]	0.1374[0.0253]
		II	0.1004[0.0203]	0.1603[0.0353]	0.1455[0.0299]	0.0964[0.0159]	0.1324[0.0307]	0.1236[0.0224]
		III	0.1137[0.0224]	0.1574[0.0395]	0.1656[0.0369]	0.1010[0.0144]	0.1279[0.0298]	0.1401[0.0288]

Table 6
CPs and AWs (within bracket) for model parameters with $(\alpha_1, \alpha_2, \beta) = (2.3, 1.9, 1)$.

n	m	CS	ACI			BCI		
			α_1	α_2	β	α_1	α_2	β
30	15	I	0.9190[0.7237]	0.9223[0.7765]	0.9244[0.9232]	0.9163[0.6533]	0.9184[0.7235]	0.9225[0.8543]
		II	0.9179[0.7045]	0.9214[0.7689]	0.9231[0.8946]	0.9172[0.6680]	0.9201[0.7462]	0.9213[0.8462]
		III	0.9184[0.7169]	0.9200[0.7593]	0.9245[0.9053]	0.9169[0.6645]	0.9195[0.7351]	0.9234[0.8519]
	25	I	0.9245[0.5460]	0.9276[0.6324]	0.9310[0.7347]	0.9231[0.5019]	0.9258[0.5846]	0.9290[0.6834]
		II	0.9236[0.5326]	0.9290[0.6411]	0.9297[0.7250]	0.9218[0.4938]	0.9247[0.5630]	0.9286[0.6642]
		III	0.9238[0.5412]	0.9281[0.6370]	0.9308[0.7305]	0.9226[0.4892]	0.9260[0.5762]	0.9305[0.6901]
60	30	I	0.9330[0.3328]	0.9358[0.4509]	0.9369[0.5361]	0.9304[0.2841]	0.9334[0.3689]	0.9341[0.4365]
		II	0.9322[0.3259]	0.9342[0.4338]	0.9351[0.5300]	0.9315[0.2932]	0.9335[0.3724]	0.9356[0.4587]
		III	0.9329[0.3244]	0.9356[0.4421]	0.9382[0.5248]	0.9322[0.3015]	0.9343[0.3760]	0.9364[0.4590]
	45	I	0.9385[0.2578]	0.9423[0.3230]	0.9441[0.3752]	0.9376[0.2174]	0.9401[0.2791]	0.9429[0.3012]
		II	0.9407[0.2642]	0.9412[0.3024]	0.9439[0.3541]	0.9358[0.2169]	0.9387[0.2835]	0.9430[0.3154]
		III	0.9411[0.2672]	0.9415[0.3116]	0.9450[0.3539]	0.9392[0.2286]	0.9412[0.2874]	0.9418[0.3129]

Table 7
Complete competing risks failure data $(X_{1\cdot}, X_{2\cdot})$ for the proportion time of football game.

Origin football game proportion time					
(0.0222,0.0222)	(0.4556,0.0333)	(0.4667,0.0333)	(0.6000,0.0778)	(0.0889,0.0889)	(0.2778,0.1000)
(0.6111,0.1222)	(0.4889,0.1444)	(0.2444,0.1556)	(0.2778,0.1556)	(0.7111,0.1667)	(0.1778,0.1778)
(0.1778,0.8333)	(0.7000,0.2000)	(0.2000,0.2000)	(0.2111,0.2111)	(0.2889,0.2222)	(0.2667,0.2667)
(0.2889,0.5333)	(0.3000,0.5222)	(0.3111,0.3111)	(0.5667,0.3111)	(0.4889,0.3333)	(0.3778,0.3778)
(0.4000,0.5778)	(0.4333,0.4333)	(0.5889,0.4333)	(0.4444,0.4444)	(0.4667,0.4667)	(0.9111,0.5333)
(0.5444,0.5444)	(0.5444,0.5444)	(0.7333,0.6889)	(0.8444,0.7111)	(0.7333,0.9444)	(0.7667,0.7889)
(0.8000,0.8000)					
Compete competing risks data					
(0.0222,1)	(0.0333,2)	(0.0333,2)	(0.0778,2)	(0.0889,1)	(0.1000,2)
(0.1222,2)	(0.1444,2)	(0.1556,2)	(0.1556,2)	(0.1667,2)	(0.1778,1)
(0.1778,1)	(0.2000,2)	(0.2000,1)	(0.2111,1)	(0.2222,2)	(0.2667,1)
(0.2889,1)	(0.3000,1)	(0.3111,1)	(0.3111,2)	(0.3333,2)	(0.3778,1)
(0.4000,1)	(0.4333,1)	(0.4333,2)	(0.4444,1)	(0.4667,1)	(0.5333,2)
(0.5444,1)	(0.5444,1)	(0.6889,2)	(0.7111,2)	(0.7333,1)	(0.7667,2)
(0.8000,1)					

team scored the first kick goal. In this illustrative case, $X_{1\cdot}$ and $X_{2\cdot}$ are treated as two competing risks as cause one and cause two respectively, and the associated complete competing risk data is also provided in Table 7.

Before further proceeding, we investigate whether the Kumaraswamy distribution can be used or not to analyze these data sets. Since the Kolmogorov–Smirnov distances and the corresponding p-values (with bracket) for cause one and cause two are 0.0823(0.9558) and 0.1018(0.8176) respectively, it is seen that the Kumaraswamy distribution provides a reasonable model for these data. Moreover, as further illustration, the empirical cumulative distributions plot and overlay the theoretical power distribution, the probability–probability (P–P) and the Quantile–Quantile (Q–Q) plots are also presented under the football game proportion time shown in Fig. 1, which also imply that the Kumaraswamy distribution provides a reasonable fit. Moreover, as shown by Barreto-Souza and Lemonte [19], the null hypothesis H_0 in Test I holds and the Kumaraswamy competing risks model with common $\alpha_1 = \alpha_2 = \alpha$ could be used for fitting the football proportion data.

Based on Table 7, three sets of progressively Type-II censored competing risks sample are generated, of which the details are presented in Table 8, where since the number of failure due to cause two in origin data set is far larger than that due

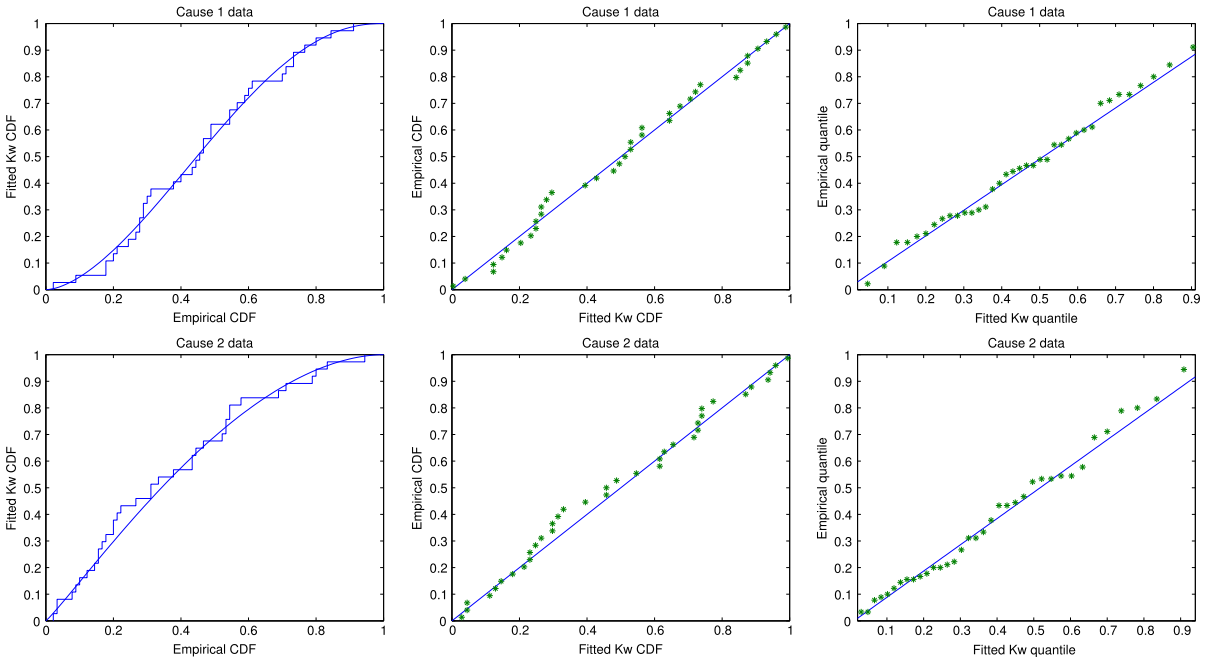


Fig. 1. Empirical distribution and fitted Kumaraswamy models, P–P and Q–Q plots at two football proportion data.

Table 8
Progressively Type-II censored competing risks data for the proportion time of football game.

Data I: $n = 37$, $m = 30$ and $R = (0^{29}, 7)$					
(0.0222,1)	(0.0333,2)	(0.0333,2)	(0.0778,2)	(0.0889,1)	(0.1000,2)
(0.1222,2)	(0.1444,2)	(0.1556,2)	(0.1556,2)	(0.1667,2)	(0.1778,1)
(0.1778,1)	(0.2000,2)	(0.2000,1)	(0.2111,1)	(0.2222,2)	(0.2667,1)
(0.2889,1)	(0.3000,1)	(0.3111,1)	(0.3111,2)	(0.3333,2)	(0.3778,1)
(0.4000,1)	(0.4333,1)	(0.4333,2)	(0.4444,1)	(0.4667,1)	(0.5333,2)
Data II: $n = 37$, $m = 30$ and $R = (7, 0^{29})$					
(0.0222,1)	(0.0333,2)	(0.0778,2)	(0.0889,1)	(0.1000,2)	(0.1222,2)
(0.1556,2)	(0.1667,2)	(0.1778,1)	(0.1778,1)	(0.2000,1)	(0.2222,2)
(0.2667,1)	(0.2889,1)	(0.3000,1)	(0.3111,1)	(0.3111,2)	(0.3778,1)
(0.4000,1)	(0.4333,1)	(0.4333,2)	(0.4444,1)	(0.4667,1)	(0.5333,2)
(0.5444,1)	(0.5444,1)	(0.6889,2)	(0.7333,1)	(0.7667,2)	(0.8000,1)
Data III: $n = 37$, $m = 30$ and $R = (1^{7}, 0^{23})$					
(0.0222,1)	(0.0333,2)	(0.0778,2)	(0.0889,1)	(0.1000,2)	(0.1222,2)
(0.1444,2)	(0.1556,2)	(0.1667,2)	(0.1778,1)	(0.1778,1)	(0.2000,2)
(0.2111,1)	(0.2222,2)	(0.2667,1)	(0.2889,1)	(0.3000,1)	(0.3111,1)
(0.3333,2)	(0.3778,1)	(0.4000,1)	(0.4333,1)	(0.4444,1)	(0.4667,1)
(0.5333,2)	(0.5444,1)	(0.5444,1)	(0.6889,2)	(0.7111,2)	(0.7333,1)

to cause one, we labeled the cause of failure is cause one in the data with equal failure times for the sake of balancing the numbers of two causes of failure. The corresponding estimates are provided in Table 9, where since we do not have any prior information about the unknown parameters, the Bayesian estimates are also obtained under almost non-informative prior with all hyper-parameters being 0.001, and the interval lengths are also provided in square brackets.

Based on results listed in Table 9, it is observed that the performance of MLEs and Bayes estimates is similar under three competing risks data sets, and that the BCIs feature shorter interval lengths than those of associated ACIs, which coincides with previous simulated results.

7. Concluding remarks

In this paper, inference is considered for the Kumaraswamy competing risks model when progressively Type-II censored samples are available. Point and interval estimates for unknown parameters are proposed under frequentist and Bayesian procedures, when the latent lifetime distributions share different and common model parameters. Real-life example and

Table 9
Point and interval estimates of β_1, β_1 and α for real applications data.

Data I	β_1	β_2	α
MLE	1.4189	1.4189	1.2402
Bayes	1.3984	1.1623	1.2113
ACI	(0.3784,2.4595)[2.0811]	(0.3784,2.4595)[2.0811]	(0.7761,1.7043)[0.9281]
HPD	(0.4131,2.3926)[1.9795]	(0.3521,2.1873)[1.8352]	(0.8124,1.6797)[0.8673]
Data II	β_1	β_2	α
MLE	1.4890	1.1387	1.3065
Bayes	1.3672	1.0732	1.3111
ACI	(0.5320,2.4460)[1.9140]	(0.3476,1.9297)[1.5820]	(0.8281,1.7849)[0.9568]
HPD	(0.4957,2.3865)[1.8908]	(0.3983,1.8926)[1.4943]	(0.7542,1.6419)[0.8877]
Data III	β_1	β_2	α
MLE	1.9744	1.5098	1.3964
Bayes	1.7210	1.2335	1.2765
ACI	(0.6478,2.8010)[2.1532]	(0.4200,2.5996)[2.1796]	(0.9230,1.8698)[0.9469]
HPD	(0.6124,2.5964)[1.9840]	(0.3953,2.0321)[1.6368]	(0.8646,1.7221)[0.8575]

simulation studies indicate that both classic and Bayesian procedures feature well behavior where Bayes results are superior to conventional classic estimates. It may be mentioned that although the failure of units due to two competing risks is considered under progressive Type-II censoring, the results can be extended to the cases with multiple causes of failure and some other failure data cases such as complete data, Type-II censoring and progressive first-failure censoring. As further work, inference for problems like optimal design and sampling plan of progressive censoring under competing risks model seem of interest, which will be studied in the future.

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Appendix A. The proof of Theorem 1

Let $m_k \neq 0$ and $\alpha_k > 0$ be fixed for $k = 1, 2$. Using the inequality $\ln t \leq t - 1$, $t > 0$ for $t = \frac{\beta_k}{\dot{\beta}_k}$, one has

$$\begin{aligned} m_k \ln \beta_k &= m_k \ln\left(\frac{\beta_k}{\dot{\beta}_k}\right) + m_k \ln \dot{\beta}_k \leq m_k \frac{\beta_k}{\dot{\beta}_k} - m_k + m_k \ln \dot{\beta}_k \\ &= \beta_k \left[-\sum_{i=1}^{m} (r_i + 1) \ln(1 - x_{i:m:n}^{\alpha_k}) \right] - m_k + m_k \ln \dot{\beta}_k, \end{aligned}$$

which implies that

$$\begin{aligned} \ell(\beta_1, \beta_2, \alpha_1, \alpha_2) &= \sum_{k=1}^{2} m_k \ln \alpha_k + \sum_{k=1}^{2} m_k \ln \beta_k + \sum_{k=1}^{2} \sum_{i=1}^{m} \alpha_k \mathbb{1}(\delta_i = k) \ln x_{i:m:n} \\ &+ \sum_{k=1}^{2} \sum_{i=1}^{m} [(r_i + 1)\beta_k - \mathbb{1}(\delta_i = k)] \ln(1 - x_{i:m:n}^{\alpha_k}) + C \\ &\leq \sum_{k=1}^{2} m_k \ln \alpha_k + \sum_{k=1}^{2} \left(\beta_k \left[-\sum_{i=1}^{m} (r_i + 1) \ln(1 - x_{i:m:n}^{\alpha_k}) \right] - m_k + m_k \ln \dot{\beta}_k \right) \\ &+ \sum_{k=1}^{2} \sum_{i=1}^{m} \alpha_k \mathbb{1}(\delta_i = k) \ln x_{i:m:n} + \sum_{k=1}^{2} \sum_{i=1}^{m} [(r_i + 1)\beta_k - \mathbb{1}(\delta_i = k)] \ln(1 - x_{i:m:n}^{\alpha_k}) + C \\ &= \sum_{k=1}^{2} m_k \ln \alpha_k + \sum_{k=1}^{2} m_k \ln \dot{\beta}_k + \sum_{k=1}^{2} \sum_{i=1}^{m} \alpha_k \mathbb{1}(\delta_i = k) \ln x_{i:m:n} \\ &- \sum_{k=1}^{2} m_k - \sum_{k=1}^{2} \sum_{i=1}^{m} \mathbb{1}(\delta_i = k) \ln(1 - x_{i:m:n}^{\alpha_k}) + C. \end{aligned}$$

Using

$$m_k = -\sum_{i=1}^{m} \dot{\beta}_k (r_i + 1) \ln(1 - x_{i:m:n}^{\alpha_k}),$$

one further has

$$\begin{aligned}\ell(\beta_1, \beta_2, \alpha_1, \alpha_2) &\le \sum_{k=1}^{2} m_k \ln \alpha_k + \sum_{k=1}^{2} m_k \ln \dot{\beta}_k + \sum_{k=1}^{2}\sum_{i=1}^{m} \alpha_k \mathbb{1}(\delta_i = k) \ln x_{i:m:n} \\ &\quad + \sum_{k=1}^{2}\sum_{i=1}^{m} [(r_i + 1)\dot{\beta}_k - \mathbb{1}(\delta_i = k)] \ln(1 - x_{i:m:n}^{\alpha_k}) + C \\ &= \ell(\dot{\beta}_1, \dot{\beta}_2, \alpha_1, \alpha_2).\end{aligned}$$

Equality holds iff $\beta_1 = \dot{\beta}_1$ and $\beta_2 = \dot{\beta}_2$. This proves the assertion. □

Appendix B. The proof of Theorem 2

Substituting $\dot{\beta}_1$ and $\dot{\beta}_2$ to (5) and ignoring the additive constant, one can obtain the profile log-likelihood function of α, say $\ell(\alpha_1, \alpha_2)$, as

$$\begin{aligned}\ell(\alpha_1, \alpha_2) &= \sum_{k=1}^{2} m_k \ln \alpha_k - \sum_{k=1}^{2} m_k \ln \left[-\sum_{i=1}^{m} (r_i + 1) \ln(1 - x_{i:m:n}^{\alpha_k}) \right] \\ &\quad - \sum_{k=1}^{2}\sum_{i=1}^{m} \mathbb{1}(\delta_i = k) \ln(1 - x_{i:m:n}^{\alpha_k}) + \sum_{k=1}^{2}\sum_{i=1}^{m} \alpha_k \mathbb{1}(\delta_i = k) \ln x_{i:m:n}.\end{aligned}$$

Taking derivative for $\ell(\alpha_1, \alpha_2)$ with respect to α_k, $k = 1, 2$, one can observe that the MLE of α_k can be obtained from (6).

In the following, the existence and uniqueness for the MLEs of α_1 and α_2 are established.

For $k = 1, 2$, since

$$\lim_{\alpha_k \to 0^+} \frac{m_k}{\alpha_k} = +\infty, \quad \lim_{\alpha_k \to 0^+} \dot{H}_1(\alpha_k) = -1, \quad \lim_{\alpha_k \to 0^+} \frac{\dot{H}_2(\alpha_k)}{\dot{H}_3(\alpha_k)} = +\infty,$$

one has $\lim_{\alpha_k \to 0^+} \dot{H}(\alpha_k) = +\infty$. Meanwhile, from

$$\lim_{\alpha_k \to +\infty} \frac{m_k}{\alpha_k} = 0, \quad \lim_{\alpha_k \to +\infty} \frac{m_k}{\alpha_k} \dot{H}_1(\alpha_k) = \sum_{i=1}^{m} \mathbb{1}(\delta_i = k) \ln x_{i:m:n},$$

$$\lim_{\alpha_k \to +\infty} \frac{m_k}{\alpha_k} \frac{\dot{H}_2(\alpha_k)}{\dot{H}_3(\alpha_k)} = -m_k \ln x_{m:m:n},$$

it is seen that $\lim_{\alpha_k \to +\infty} \dot{H}(\alpha_k) = \sum_{i=1}^{m} \mathbb{1}(\delta_i = k) \ln(x_{i:m:n}/x_{m:m:n}) < 0$.

Furthermore, taking derivative for $\dot{H}(\alpha_k)$ with respect to α_k, one has

$$\frac{d\dot{H}(\alpha_k)}{d\alpha_k} = -\frac{m_k}{\alpha_k^2} \left[D_{k1}(\alpha_k) - \frac{D_{k2}(\alpha_k)}{[\sum_{i=1}^{m} (r_i + 1) \ln(1 - x_{i:m:n}^{\alpha_k})]^2} \right],$$

with

$$D_{k1}(\alpha_k) = 1 - \frac{1}{m_k} \sum_{i=1}^{m} \frac{\mathbb{1}(\delta_i = k) x_{i:m:n}^{\alpha_k} \ln^2 x_{i:m:n}^{\alpha_k}}{(1 - x_{i:m:n}^{\alpha_k})^2},$$

$$\begin{aligned}D_{k2}(\alpha_k) &= [\sum_{i=1}^{m} (r_i + 1) \frac{x_{i:m:n}^{\alpha_k} \ln^2 x_{i:m:n}^{\alpha_k}}{[1 - x_{i:m:n}^{\alpha_k}]^2}][\sum_{i=1}^{m} (r_i + 1) \ln(1 - x_{i:m:n}^{\alpha_k})] \\ &\quad + [\sum_{i=1}^{m} (r_i + 1) \frac{x_{i:m:n}^{\alpha_k} \ln x_{i:m:n}^{\alpha_k}}{1 - x_{i:m:n}^{\alpha_k}}]^2.\end{aligned}$$

Furthermore, since function $y = \frac{t \ln^2 t}{(1-t)^2}$ increases on $(0, 1)$, then

$$D_{k1}(\alpha_k) > 1 - \frac{1}{m_k} \sum_{i=1}^{m} \mathbb{1}(\delta_i = k) \lim_{t \to 1} \frac{t \ln^2 t}{(1 - t)^2} = 0,$$

and

$$D_{k2}(\alpha_k) < [\sum_{i=1}^{m}(r_i+1)\frac{x_{i:m:n}^{\alpha_k}\ln x_{i:m:n}^{\alpha_k}}{1-x_{i:m:n}^{\alpha_k}}]^2 - [\sum_{i=1}^{m}(r_i+1)\frac{x_{i:m:n}^{\alpha_k}\ln^2 x_{i:m:n}^{\alpha_k}}{[1-x_{i:m:n}^{\alpha_k}]^2}][\sum_{i=1}^{m}(r_i+1)x_{i:m:n}^{\alpha_k}] < 0,$$

where the last inequality is obtained by using Cauchy–Schwarz inequality. Therefore, one has $\frac{d\dot{H}(\alpha_k)}{d\alpha_k} < 0$ which implies that $\dot{H}(\alpha_k)$ is a decreasing function from positive to negative. Hence, the result is proved. □

Appendix C. The proof of Theorem 3

When $\alpha_1 = \alpha_2 = \alpha$, the log-likelihood function of β_1, β_2 and α, say $\ell(\beta_1, \beta_2, \alpha)$, can be derived from (5) as

$$\ell(\beta_1,\beta_2,\alpha) = m\ln\alpha + \alpha\sum_{i=1}^{m}\ln x_i + \sum_{k=1}^{2} m_k\ln\beta_k + \sum_{i=1}^{m}[(\beta_1+\beta_2)(r_i+1)-1]\ln(1-x_{i:m:n}^{\alpha}) + C.$$

Similar as previous, one has

$$m_k\ln\beta_k \le \beta_k\left[-\sum_{i=1}^{m}(r_i+1)\ln(1-x_{i:m:n}^{\alpha})\right] - m_k + m_k\ln\ddot{\beta}_k.$$

Therefore,

$$\begin{aligned}\ell(\beta_1,\beta_2,\alpha) &\le m\ln\alpha + \alpha\sum_{i=1}^{m}\ln x_{i:m:n} + \sum_{k=1}^{2}\left[\beta_k\left[-\sum_{i=1}^{m}(r_i+1)\ln(1-x_{i:m:n}^{\alpha})\right] - m_k + m_k\ln\ddot{\beta}_k\right] \\ &\quad + \sum_{i=1}^{m}[(\beta_1+\beta_2)(r_i+1)-1]\ln(1-x_{i:m:n}^{\alpha}) + C \\ &= m\ln\alpha + \alpha\sum_{i=1}^{m}\ln x_i + \sum_{k=1}^{2} m_k\ln\ddot{\beta}_k - m - \sum_{i=1}^{m}\ln(1-x_{i:m:n}^{\alpha}) + C,\end{aligned}$$

using

$$\sum_{i=1}^{m}[(\ddot{\beta}_1+\ddot{\beta}_2)(r_i+1)-1]\ln(1-x_{i:m:n}^{\alpha}) = -\sum_{i=1}^{m}\ln(1-x_{i:m:n}^{\alpha}) - m,$$

one further has

$$\ell(\beta_1,\beta_2,\alpha) \le m\ln\alpha + \alpha\sum_{i=1}^{m}\ln x_{i:m:n} + \sum_{k=1}^{2} m_k\ln\ddot{\beta}_k + \sum_{i=1}^{m}[(\ddot{\beta}_1+\ddot{\beta}_2)(r_i+1)-1]\ln(1-x_{i:m:n}^{\alpha}) + C = \ell(\ddot{\beta}_1,\ddot{\beta}_2,\alpha).$$

Equality holds iff $\beta_1 = \ddot{\beta}_1$ and $\beta_2 = \ddot{\beta}_2$. This proves the assertion. □

Appendix D. The proof of Theorem 4

Substituting $\ddot{\beta}_1$ and $\ddot{\beta}_2$ to the log-likelihood function $\ell(\beta_1, \beta_2, \alpha)$ shown in Appendix C and ignoring the additive constant, the profile log-likelihood function of α, say $\ell(\alpha)$, is

$$\ell(\alpha) = m\ln\alpha + \sum_{i=1}^{m}\ln\left[\frac{x_{i:m:n}^{\alpha}}{1-x_{i:m:n}^{\alpha}}\right] - \ln[-\sum_{i=1}^{m}(r_i+1)\ln(1-x_{i:m:n}^{\alpha})].$$

By taking derivative for $\ell(\alpha)$ with respect to α, it is seen that the MLE of α can be derived from (8).

Since $\lim_{\alpha\to 0^+}\frac{m}{\alpha}=+\infty$, $\lim_{\alpha\to 0^+}\ddot{H}_1(\alpha)=-1$, $\lim_{\alpha\to 0^+}\frac{\ddot{H}_2(\alpha)}{\ddot{H}_3(\alpha)}=+\infty$, one has $\lim_{\alpha\to 0^+}\ddot{H}(\alpha)=+\infty$. Meanwhile, from $\lim_{\alpha\to+\infty}\frac{m}{\alpha}=0$, $\lim_{\alpha\to+\infty}\frac{m}{\alpha}\ddot{H}_1(\alpha)=\sum_{i=1}^{m}\ln x_{i:m:n}$ and $\lim_{\alpha\to+\infty}\frac{m}{\alpha}\frac{\ddot{H}_2(\alpha)}{\ddot{H}_3(\alpha)}=-m\ln x_{m:m:n}$, it is seen that $\lim_{\alpha\to+\infty}\ddot{H}(\alpha)=\sum_{i=1}^{m}\ln(x_{i:m:n}/x_{m:m:n})<0$.

Furthermore, since

$$\frac{d\ddot{H}(\alpha)}{d\alpha}=-\frac{m}{\alpha^2}\left[D_1(\alpha)-\frac{D_2(\alpha)}{[\sum_{i=1}^{m}(r_i+1)\ln(1-x_{i:m:n}^{\alpha})]^2}\right],$$

with

$$D_1(\alpha)=1-\frac{1}{m}\sum_{i=1}^{m}\frac{x_{i:m:n}^{\alpha}\ln^2 x_{i:m:n}^{\alpha}}{(1-x_{i:m:n}^{\alpha})^2}>0,$$

and

$$\begin{aligned}D_2(\alpha)=&\left(\sum_{i=1}^{m}(r_i+1)\frac{x_{i:m:n}^{\alpha}\ln^2 x_{i:m:n}^{\alpha}}{(1-x_{i:m:n}^{\alpha})^2}\right)\left(\sum_{i=1}^{m}(r_i+1)\ln(1-x_{i:m:n}^{\alpha})\right)\\&+\left(\sum_{i=1}^{m}(r_i+1)\frac{x_{i:m:n}^{\alpha}\ln x_{i:m:n}^{\alpha}}{1-x_{i:m:n}^{\alpha}}\right)^2<0,\end{aligned}$$

then function $\ddot{H}(\alpha)$ decreases from positive to negative. Therefore, the result is proved. □

Appendix E. The proof of Theorem 5

From (5), the log-likelihood function of α_1, α_2 and β, say $\ell(\alpha_1,\alpha_2,\beta)$, is

$$\begin{aligned}\ell(\alpha_1,\alpha_2,\beta)=&\sum_{k=1}^{2}m_k\ln\alpha_k+m\ln\beta+\sum_{k=1}^{2}\sum_{i=1}^{m}\alpha_k\mathbb{1}(\delta_i=k)\ln x_{i:m:n}\\&+\sum_{k=1}^{2}\sum_{i=1}^{m}[(r_i+1)\beta-\mathbb{1}(\delta_i=k)]\ln(1-x_{i:m:n}^{\alpha_k})+C.\end{aligned}$$

Since

$$m\ln\beta\le\beta\left[-\sum_{k=1}^{2}\sum_{i=1}^{m}(r_i+1)\ln(1-x_{i:m:n}^{\alpha_k})\right]-m+m\ln\dddot{\beta},$$

then

$$\begin{aligned}&\ell(\beta,\alpha_1,\alpha_2)\\&\quad\le\sum_{k=1}^{2}m_k\ln\alpha_k+\beta\left[-\sum_{k=1}^{2}\sum_{i=1}^{m}(r_i+1)\ln(1-x_{i:m:n}^{\alpha_k})\right]-m+m\ln\ddot{\beta}\\&\quad+\sum_{k=1}^{2}\sum_{i=1}^{m}\alpha_k\mathbb{1}(\delta_i=k)\ln x_{i:m:n}\\&\quad+\sum_{k=1}^{2}\sum_{i=1}^{m}[(r_i+1)\beta-\mathbb{1}(\delta_i=k)]\ln(1-x_{i:m:n}^{\alpha_k})+C\\&\quad=\sum_{k=1}^{2}m_k\ln\alpha_k+m\ln\ddot{\beta}+\sum_{k=1}^{2}\sum_{i=1}^{m}\alpha_k\mathbb{1}(\delta_i=k)\ln x_{i:m:n}\\&\quad-m-\sum_{k=1}^{2}\sum_{i=1}^{m}\mathbb{1}(\delta_i=k)\ln(1-x_{i:m:n}^{\alpha_k})+C,\end{aligned}$$

using

$$m=-\sum_{k=1}^{2}\sum_{i=1}^{m}(r_i+1)\ddot{\beta}\ln(1-x_{i:m:n}^{\alpha_k}),$$

one further has

$$\ell(\alpha_1, \alpha_2, \beta) = \sum_{k=1}^{2} m_k \ln \alpha_k + m \ln \dddot{\beta} + \sum_{k=1}^{2}\sum_{i=1}^{m} \alpha_k \mathbb{1}(\delta_i = k) \ln x_{i:m:n} + \sum_{k=1}^{2}\sum_{i=1}^{m} [(r_i + 1)\dddot{\beta} - \mathbb{1}(\delta_i = k)] \ln(1 - x_{i:m:n}^{\alpha_k}) + C,$$

Equality holds iff $\beta = \dddot{\beta}$. This proves the assertion. □

Appendix F. The elements of fisher information

$$I_{11} = \frac{\partial^2 \ell(\beta_1, \beta_2, \alpha_1, \alpha_2)}{\partial \beta_1^2} = -\frac{m_1}{\beta_1^2},\ I_{12} = \frac{\partial^2 \ell(\beta_1, \beta_2, \alpha_1, \alpha_2)}{\partial \beta_1 \partial \beta_2} = 0,$$

$$I_{13} = \frac{\partial^2 \ell(\beta_1, \beta_2, \alpha_1, \alpha_2)}{\partial \beta_1 \partial \alpha_1} = -\sum_{i=1}^{m} (r_i + 1) \frac{x_{i:m:n}^{\alpha_1} \ln x_{i:m:n}}{1 - x_{i:m:n}^{\alpha_1}},$$

$$I_{14} = \frac{\partial^2 \ell(\beta_1, \beta_2, \alpha_1, \alpha_2)}{\partial \beta_1 \partial \alpha_2} = 0,\ I_{22} = \frac{\partial^2 \ell(\beta_1, \beta_2, \alpha_1, \alpha_2)}{\partial \beta_2^2} = -\frac{m_2}{\beta_2^2},$$

$$I_{23} = \frac{\partial^2 \ell(\beta_1, \beta_2, \alpha_1, \alpha_2)}{\partial \beta_2 \partial \alpha_1} = 0,$$

$$I_{24} = \frac{\partial^2 \ell(\beta_1, \beta_2, \alpha_1, \alpha_2)}{\partial \beta_2 \partial \alpha_2} = -\sum_{i=1}^{m} (r_i + 1) \frac{x_{i:m:n}^{\alpha_2} \ln x_{i:m:n}}{1 - x_{i:m:n}^{\alpha_2}},$$

$$I_{33} = \frac{\partial^2 \ell(\beta_1, \beta_2, \alpha_1, \alpha_2)}{\partial \alpha_1^2} = -\frac{m_1}{\alpha_1^2} - \sum_{i=1}^{m} [(r_i + 1)\beta_1 - \mathbb{1}(\delta_i = 1)] \frac{x_{i:m:n}^{\alpha_1} \ln^2 x_{i:m:n}}{[1 - x_{i:m:n}^{\alpha_1}]^2}$$

$$I_{34} = \frac{\partial^2 \ell(\beta_1, \beta_2, \alpha_1, \alpha_2)}{\partial \alpha_1 \partial \alpha_2} = 0,$$

$$I_{44} = \frac{\partial^2 \ell(\beta_1, \beta_2, \alpha_1, \alpha_2)}{\partial \alpha_2^2} = -\frac{m_2}{\alpha_2^2} - \sum_{i=1}^{m} [(r_i + 1)\beta_2 - \mathbb{1}(\delta_i = 2)] \frac{x_{i:m:n}^{\alpha_2} \ln^2 x_{i:m:n}}{[1 - x_{i:m:n}^{\alpha_2}]^2}.$$

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